

Quantification of Inter-Limb Asymmetry During Self-Feeding Tasks Using OpenSim-Based Upper Limb Kinematic Analysis

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This study quantifies inter-limb asymmetry during standardised self-feeding tasks in healthy individuals using OpenSim musculoskeletal modelling. Two participants performed four seated eating-related tasks (bottle opening, spoon use, drinking, and cutting), each repeated twice with both dominant and non-dominant hands. A generic upper-body model with two additional markers was scaled to each subject using static marker data, and inverse kinematics was performed to estimate joint angles. Movement duration, range of motion (ROM) were computed using MATLAB. Kinematic data was time-normalized (0-100%) for inter-limb comparison. Results reveal task-dependent asymmetries, particularly in the shoulder abduction/adduction ROM.

1 Introduction

Activities of Daily Living (ADLs) are fundamental tasks that enable individuals to maintain independence and quality of life. Among these activities, self-feeding represents one of the most essential functions, requiring coordinated interaction between the trunk, arm, forearm, and hand. Successful execution of eating-related movements depends on complex motor control strategies involving joint coordination, movement smoothness, and postural stability.

Upper limb impairments are common in several neurological conditions, particularly following stroke. Motor deficits often lead to altered movement patterns, reduced joint mobility, and compensatory trunk motions that negatively affect functional performance. Consequently, quantitative assessment of upper limb kinematics has become an important tool in rehabilitation research [1], allowing researchers to evaluate motor impairments and monitor recovery.

Advances in motion capture technologies and musculoskeletal modelling platforms such as OpenSim have enabled more precise biomechanical analyses of upper limb movements. By combining experimental marker trajectories with subject-specific musculoskeletal mod-

els, it is possible to estimate joint kinematics during functional tasks. This approach provides valuable insight into movement strategies that may not be detectable through conventional clinical assessments alone.

Although numerous studies have investigated upper limb dysfunction in neurological populations [2], recently not that much attention has been paid to the natural asymmetries that exist between dominant and non-dominant limbs in healthy individuals. Human movement is inherently asymmetric, and differences in motor control, strength, and coordination between limbs can influence task execution. Understanding these normal asymmetry patterns is essential before attempting to identify pathological deviations.

Self-feeding tasks are particularly suitable for studying inter-limb asymmetry because they are highly relevant to daily life, involve multiple joints and degrees of freedom, and can be reproduced consistently in a laboratory environment. Furthermore, feeding activities are frequently used as outcome measures in neurorehabilitation due to their importance for functional independence.

Therefore, the aim of this study was to quantify inter-limb asymmetry during standardized self-feeding

tasks in healthy young adults using an OpenSim-based kinematic analysis framework. Joint range of motion, movement duration, and symmetry indices were computed from inverse kinematic data obtained during four representative eating-related tasks: bottle opening, spoon use, drinking from a glass, and cutting with knife and fork.

Research Question *How does the non-dominant upper limb differ biomechanically from the dominant limb during standardized self-feeding tasks in healthy individuals?*

A secondary objective was to investigate whether asymmetry patterns remain consistent across different tasks or whether task complexity influences the magnitude of biomechanical differences between limbs. The results of this study aim to establish a healthy baseline that may serve as a reference for future comparisons with post-stroke and other neurological populations.

2 Methods

2.1 Participants

Two healthy male volunteers (23 and 24 years old) participated in this study. Both participants reported no history of neurological, musculoskeletal, or upper-limb disorders that could affect movement performance. All participants were familiarised with the experimental procedure before data collection.

The objective of the study was not to establish population-level statistics but rather to evaluate the feasibility of an OpenSim-based framework for quantifying inter-limb asymmetry during self-feeding activities and to obtain preliminary baseline data from healthy individuals.

2.2 Experimental setup

Data collection was performed at the Biomechanical Engineering Laboratory (BIOMECH Lab) of ETSEIB, Universitat Politècnica de Catalunya (UPC). Participants were seated comfortably in front of a table at a

standard eating height. The experimental setup was designed to reproduce common self-feeding activities while minimising external constraints on movement execution.

Reflective markers were attached according to the selected upper-limb motion capture protocol, and subjects were instructed to perform all tasks at a self-selected and comfortable speed, similar to their natural daily-life behaviour.

2.3 Experimental tasks

Four representative self-feeding activities were selected because of their relevance to activities of daily living (ADLs) and due to some of them being of frequent use in rehabilitation assessment [3]:

- **Bottle opening:** opening and closing a bottle using one hand.
- **Spoon task:** transporting a spoon from a bowl to the mouth.
- **Drinking task:** lifting a glass and bringing it to the mouth.
- **Cutting task:** simulating food cutting using a knife and fork.

Each task was performed using both the dominant (D) and non-dominant (ND) upper limbs. Two repetitions per limb were recorded for each task, resulting in a total of 16 dynamic trials per participant.

The start and end positions were standardised as much as possible to improve repeatability between repetitions. The chosen position was having both palms on the table.

2.4 Motion capture marker protocol

Upper-limb kinematics were recorded using Protocol B, consisting of 18 reflective markers placed on anatomical landmarks of the trunk, shoulder complex, upper arm, forearm, and wrist.

To improve wrist motion tracking, two additional markers were placed on the dorsal surface of the hand, resulting in a total of 20 markers. These supplementary markers were intended to provide additional information on wrist orientation.

Marker placement was performed consistently across subjects following laboratory guidelines.

2.5 Musculoskeletal model scaling

A generic OpenSim upper-body musculoskeletal model was adapted to each participant using a static calibration trial.

The scaling procedure adjusted segment dimensions according to individual anthropometric characteristics while preserving the original joint definitions and model structure.

Scaling quality was assessed using marker reconstruction errors provided by OpenSim. The following criteria were used to evaluate scaling performance:

- Root Mean Square (RMS) marker error as small as possible.
- Iterative adjustment of marker locations and segment scaling factors.
- Visual inspection of marker alignment and realistic joint positions throughout the model.

For Subject 1, scaling errors remained below 1 cm for most markers. Subject 2 presented larger residual errors, particularly around the elbow and neck regions, reaching approximately 4 cm in some cases. Nevertheless, the resulting model provided a sufficiently accurate representation of the participant's upper-limb geometry for subsequent kinematic analysis.

2.6 Inverse Kinematics (IK)

Since no ground reaction forces or external force measurements were recorded during the experimental sessions, the biomechanical analysis was restricted to kinematic variables. Therefore, only the OpenSim IK tool

was used to estimate joint angles from the experimental marker trajectories.

IK computes the set of joint coordinates that minimizes the difference between experimentally measured marker positions and their corresponding virtual markers in the musculoskeletal model. The optimization problem solved by OpenSim can be expressed as Equation 1.

$$\min \sum_{i=1}^N w_i \left\| x_i^{exp} - x_i^{model} \right\|^2 \quad (1)$$

where x_i^{exp} represents the position of the experimental marker, x_i^{model} the corresponding model marker position, and w_i the weight assigned to marker i .

Marker weights were adjusted iteratively according to laboratory experience and marker reliability. Anatomical landmarks with higher tracking accuracy and greater biomechanical relevance received larger weights, whereas markers prone to misplacement or reduced visibility received lower weights.

The following weighting strategy was adopted:

- Higher weight: elbow, wrist, acromion and neck markers.
- Lower weight: posterior pelvis and T10 markers.

Several IK iterations were performed until a final solution was selected based on RMS error and maximum marker error (around 4 cm or less in the majority of cases), visual inspection of the reconstructed motion, and consistency of the resulting joint angle trajectories.

2.7 Data pre-processing

Following IK, the resulting joint angle trajectories were exported from OpenSim and prepared for subsequent analysis.

For each task, movement cycles were manually identified based on the beginning and end of the functional movement. Joint angle trajectories corresponding to the dominant and non-dominant limbs were then segmented accordingly.

The processed trajectories were subsequently exported to MATLAB for quantitative analysis of movement duration, range of motion, and inter-limb symmetry.

2.8 MATLAB post-processing

The joint angle trajectories obtained from OpenSim were exported and processed in MATLAB. Custom scripts were developed to extract, normalise, and analyse the kinematic variables of interest for each participant and task.

To reduce trial-to-trial variability, the two repetitions belonging to the same task and limb were averaged prior to the calculation of metrics. This approach provided a representative kinematic profile for each participant dominant and non-dominant limbs and task.

2.8.1 Time normalization

Because movement duration varied between repetitions and participants, all trials were time-normalized to a common scale ranging from 0% to 100% of task completion.

This procedure enabled direct comparison of joint kinematic profiles between dominant and non-dominant limbs independently of execution speed.

2.8.2 Movement duration

Movement duration was defined as the time elapsed between the start and end of each task cycle.

$$T = t_{final} - t_{initial} \quad (2)$$

where $t_{initial}$ and t_{final} correspond to the beginning and end of the movement, respectively.

For each participant, the average movement duration was computed across repetitions for every task and limb. This metric was used to evaluate potential temporal asymmetries between dominant and non-dominant upper-limb performance.

2.8.3 Range of Motion (ROM)

ROM was used to quantify the angular excursion of each degree of freedom during task execution. ROM represents the difference between the maximum and minimum joint angles reached throughout a movement cycle and provides an indication of the extent of joint utilization during task performance.

$$ROM = \theta_{max} - \theta_{min} \quad (3)$$

where θ_{max} and θ_{min} are the maximum and minimum joint angles observed during the movement cycle, respectively.

ROM was computed for the following degrees of freedom (DOFs), which were the most significant after the data pre-processing.

- Shoulder flexion/extension
- Shoulder abduction/adduction
- Elbow flexion/extension
- Forearm pronation/supination

These DOFs were also selected because they represent the primary joint motions involved in self-feeding activities and are directly related to reaching, object manipulation, and food-to-mouth movements.

For each participant and task, ROM values were calculated separately for the dominant and non-dominant limbs. These values were subsequently used to assess inter-limb differences and compute symmetry indices.

2.8.4 Symmetry Index (SI)

Inter-limb asymmetry was quantified using the SI described by Equation 4 [4], which expresses the relative difference between dominant and non-dominant limb performance:

$$SI = \frac{|X_D - X_{ND}|}{0.5(X_D + X_{ND})} \times 100 \quad (4)$$

where X_D and X_{ND} represent the metric obtained for the dominant and non-dominant limbs, respectively.

An SI value of 0% indicates perfect symmetry, whereas larger values indicate increasing asymmetry between limbs. This normalization allows comparison between variables with different magnitudes and units.

The SI was calculated for each degree of freedom and task using the corresponding ROM values obtained from the dominant and non-dominant limbs.

3 Results

3.1 Movement duration

Average movement duration was calculated for each task and limb to evaluate potential temporal asymmetries during task execution. To compare the duration differences between dominant and non-dominant limbs the SI measure in % was also adopted.

Overall, differences in movement duration between dominant and non-dominant limbs were relatively small for the bottle opening, spoon, and drinking tasks. The lowest asymmetry values were observed during drinking, with Symmetry Index values of 10.7% for Subject 1 and 3.4% for Subject 2.

In contrast, the cutting task exhibited the largest temporal asymmetries in both participants. Subject 1 presented a duration Symmetry Index of 27.6%, whereas Subject 2 reached 42.2%, indicating substantial differences in execution time between limbs.

Interestingly, larger asymmetries in movement duration were not always associated with larger differences in ROM, suggesting that temporal and kinematic asymmetries may reflect different movement strategies.

These findings indicate that task complexity has a greater influence on temporal asymmetry than simple unilateral reaching movements, with cutting representing the most demanding and asymmetric task among those evaluated.

3.2 Range of motion

3.2.1 Bottle opening and spoon tasks

ROM values were calculated for all tasks and degrees of freedom in both participants. Table 1 summarizes the dominant and non-dominant ROM values together with the corresponding Symmetry Index (SI) for the bottle opening and spoon tasks.

Overall, Subject 1 exhibited lower inter-limb asymmetry than Subject 2 across most tasks and degrees of freedom. Shoulder abduction/adduction consistently showed the largest asymmetry values, reaching SI values of 51.9% and 54.3% during the bottle opening task for Subjects 1 and 2, respectively.

In contrast, shoulder flexion/extension during the spoon task displayed nearly symmetrical behaviour in both participants, with SI values of 0.2% for Subject 1 and 3.6% for Subject 2. Similarly, elbow flexion/extension during the spoon task showed almost identical ROM values between limbs in Subject 1 (SI = 1.0%).

Subject 2 generally presented larger asymmetries than Subject 1, particularly for elbow flexion/extension during both bottle opening (SI = 30.6%) and spoon use (SI = 30.9%). These findings suggest greater variability in movement strategies despite both individuals being healthy and free from neurological impairment.

The results indicate that shoulder abduction/adduction was the most variable degree of freedom across participants, whereas shoulder flexion/extension exhibited the most consistent movement patterns between limbs.

Figure 1 illustrates the time-normalized joint angle trajectories during the bottle opening task for both participants. Dominant and non-dominant limb movements are represented throughout the normalized movement cycle (0–100%).

Visual inspection of the kinematic profiles supports the ROM findings reported in Table 1. The largest differences between limbs are observed in shoulder abduction/adduction, particularly for Subject 2, where the

Table 1: Range of Motion (ROM) and Symmetry Index (SI) for bottle opening and spoon tasks.

Task	Task DOF	S1 D-ROM	S1 ND-ROM	S1 SI (%)	S2 D-ROM	S2 ND-ROM	S2 SI (%)
Bottle	Shoulder Abd/Add	16.7	9.8	51.9	25.5	44.6	54.3
	Shoulder Flex/Ext	82.9	71.9	14.3	82.1	76.6	7.0
	Elbow Flex/Ext	34.9	32.0	8.7	50.1	68.1	30.6
	Forearm Pro/Sup	55.8	69.1	21.2	87.3	74.1	16.3
Spoon	Shoulder Abd/Add	47.3	36.3	26.2	43.6	29.7	38.0
	Shoulder Flex/Ext	93.7	94.0	0.2	112.4	108.5	3.6
	Elbow Flex/Ext	60.0	60.6	1.0	62.2	45.6	30.9
	Forearm Pro/Sup	68.6	75.5	9.5	88.0	82.5	6.6

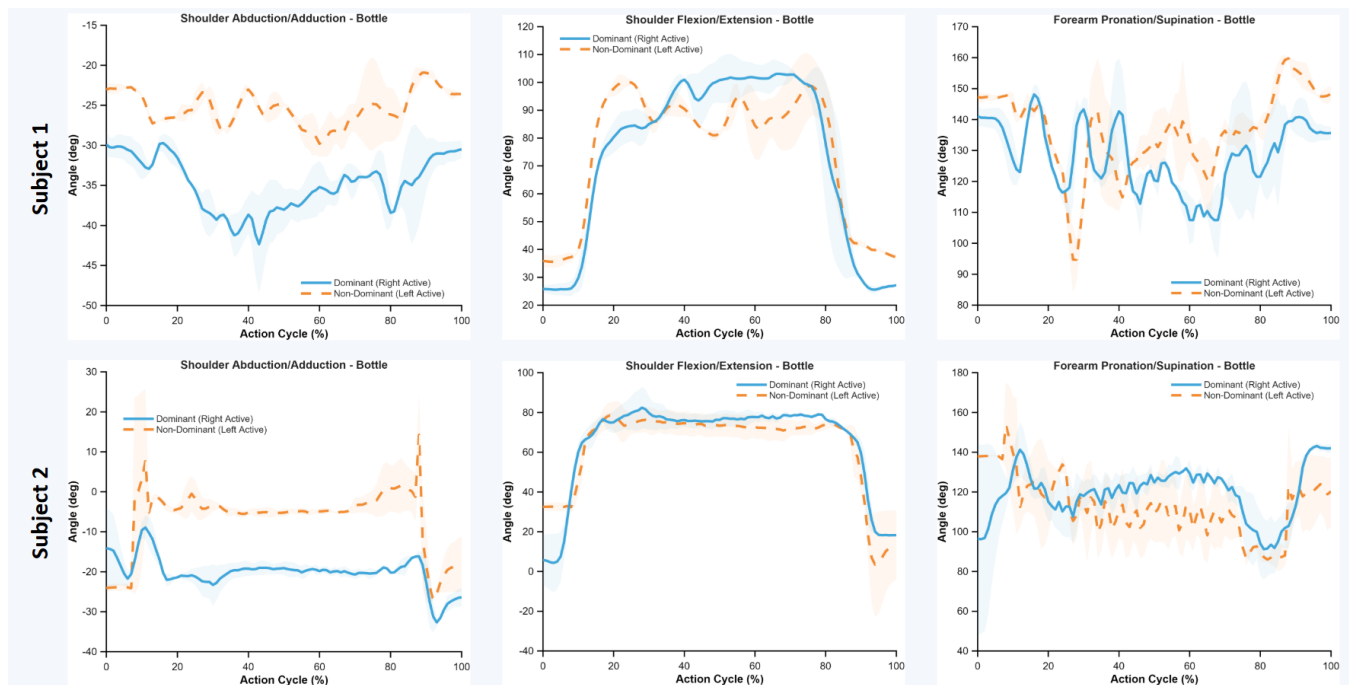


Figure 1: Time-normalised joint angle trajectories during the bottle opening task. The upper row corresponds to Subject 1 and the lower row to Subject 2. Columns represent shoulder abduction/adduction, shoulder flexion/extension, and forearm pronation/supination, respectively. Solid blue lines indicate dominant limb performance, whereas dashed orange lines represent non-dominant limb performance. Shaded regions correspond to variability across repetitions.

dominant and non-dominant trajectories remain clearly separated throughout most of the movement cycle. In contrast, shoulder flexion/extension exhibits highly similar profiles between limbs, especially during the plateau phase corresponding to the bottle manipulation period.

Forearm pronation/supination demonstrates moderate asymmetry in both subjects, reflecting the rotational demands associated with bottle opening. These results

further suggest that shoulder abduction/adduction is the most influenced DOF by individual motor strategies during this task.

3.2.2 Drinking and cutting tasks

Table 2 summarises the dominant and non-dominant ROM values together with the corresponding SI for the drinking and cutting tasks.

Compared with bottle opening and spoon use, the

cutting task produced the largest asymmetries observed in the study. In particular, Subject 2 exhibited an exceptionally high asymmetry in shoulder abduction/adduction (SI = 84.5%), indicating markedly different movement strategies between dominant and non-dominant limbs during this task.

Drinking movements generally showed lower asymmetry levels, although Subject 2 still demonstrated considerable differences in shoulder flexion/extension (SI = 37.4%). In contrast, forearm pronation/supination remained relatively symmetric in both participants during drinking, with SI values below 8%.

Elbow flexion/extension exhibited highly symmetric behaviour during cutting in Subject 1 (SI = 0.1%), whereas larger differences were observed in Subject 2 (SI = 10.9%). These findings suggest that asymmetry is strongly task-dependent and may vary substantially between individuals despite the absence of neurological impairment.

Figure 2 presents the time-normalized joint angle trajectories for the drinking and cutting tasks. The kinematic profiles confirm the ROM findings reported in Table 2, showing larger inter-limb differences during cutting than during drinking.

Visual inspection of the trajectories reveals that drinking movements are characterized by relatively similar profiles between limbs, particularly in Subject 1. In contrast, cutting movements exhibit greater divergence between dominant and non-dominant trajectories, especially in Subject 2, reflecting the elevated symmetry index values reported in Table 2. These observations further support the conclusion that task complexity plays a major role in determining inter-limb asymmetry.

Additionally, Subject 2's shoulder flexion/extension during drinking presented a 37.4% SI. In the corresponding plot in Figure 2 it is possible to see how the ROM difference is partly due to the initial position misalignment of the movement between the dominant and the non-dominant limbs. Thus, making evident the influence of the initial position in the ROM SI results.

4 Discussion

Overall, Subject 1 exhibited lower asymmetry levels than Subject 2 across most tasks and degrees of freedom. This finding highlights the considerable inter-subject variability that may exist even within a healthy population and suggests that individual motor strategies play an important role during task execution.

Shoulder abduction/adduction consistently presented the highest SI values across participants and tasks. To compensate the movement with the non-dominant hand, it has been seen that participants may try to tackle the precision of the movement by shoulder abduction in such way that they have more limb control.

In particular, the bottle opening and cutting tasks showed pronounced asymmetries, reaching SI values above 50% and 80% in some conditions. This DOF appears to be more dependent on individual movement strategies and less constrained than elbow flexion/extension, which generally exhibited lower asymmetry values.

Task complexity also influenced movement symmetry. Bottle opening and cutting tasks produced larger asymmetries than spoon and drinking tasks, likely because they require more complex coordination and object manipulation. In contrast, spoon and drinking movements followed more stereotyped trajectories and therefore tended to produce lower asymmetry values.

Interestingly, differences in movement duration were not always associated with differences in ROM. Some tasks exhibited relatively symmetric execution times while maintaining considerable ROM asymmetries. This suggests that temporal and kinematic asymmetries may reflect different aspects of motor control and should therefore be interpreted independently.

4.1 Limitations

Several limitations should be considered when interpreting the results of this study.

First, only two healthy participants were included, preventing statistical analysis and limiting the general-

Table 2: Range of Motion (ROM) and Symmetry Index (SI) for drinking and cutting tasks.

Task	Task DOF	S1 D-ROM	S1 ND-ROM	S1 SI (%)	S2 D-ROM	S2 ND-ROM	S2 SI (%)
Drink	Shoulder Abd/Add	14.3	11.7	20.2	13.1	18.1	32.4
	Shoulder Flex/Ext	73.8	65.5	12.0	58.4	40.0	37.4
	Elbow Flex/Ext	57.8	67.3	15.3	56.7	69.3	20.0
	Forearm Pro/Sup	46.3	47.8	3.1	62.4	57.9	7.5
Cutting	Shoulder Abd/Add	17.6	14.1	22.2	17.0	6.8	84.5
	Shoulder Flex/Ext	27.9	29.9	7.1	22.9	14.4	45.6
	Elbow Flex/Ext	57.9	57.9	0.1	26.4	23.7	10.9
	Forearm Pro/Sup	52.5	40.7	25.3	45.0	46.0	2.2

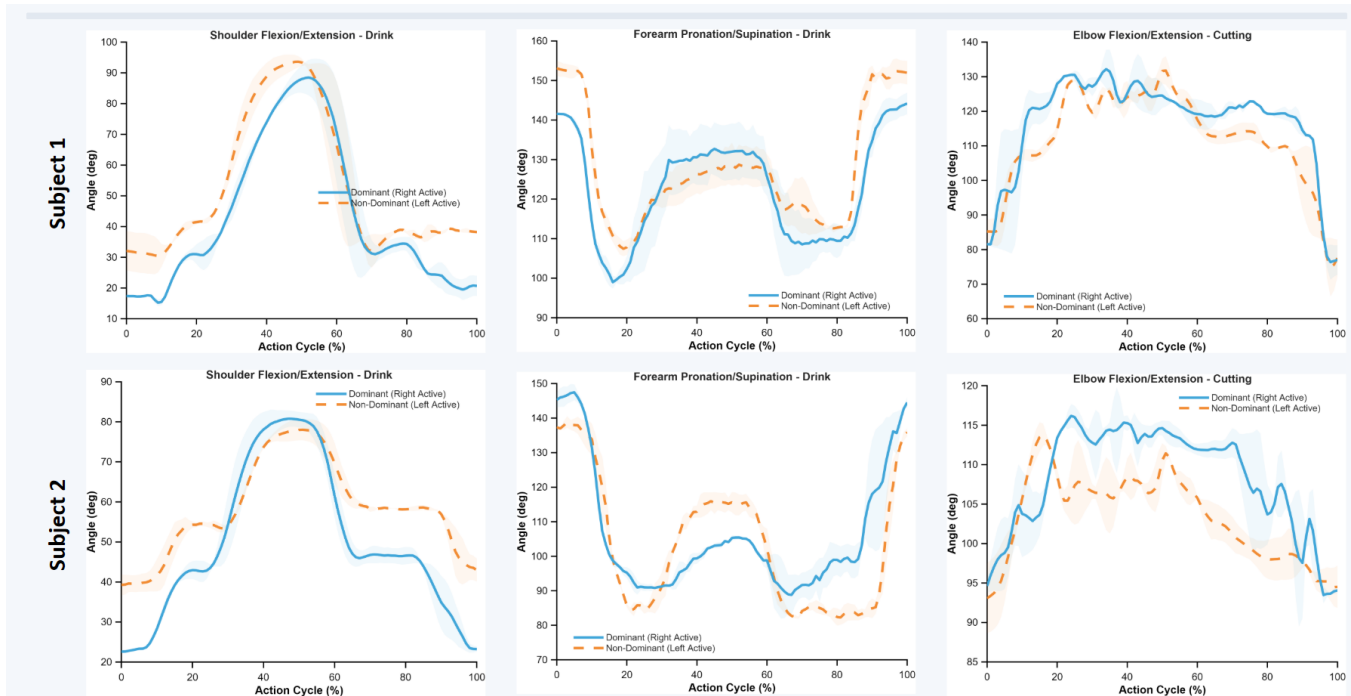


Figure 2: Time-normalised joint angle trajectories during the drinking and cutting tasks. The upper row corresponds to Subject 1 and the lower row to Subject 2. Columns represent shoulder flexion/extension during drinking, forearm pronation/supination during drinking, and elbow flexion/extension during cutting. Solid blue lines indicate dominant limb performance and dashed orange lines indicate non-dominant limb performance.

izability of the findings. Second, movement smoothness and trunk compensation were initially considered as outcome measures but were not included in the final analysis due to time constraints. Finally, variability in initial posture and task execution may have influenced some of the observed asymmetries.

Future studies should include a larger sample size, incorporate trunk compensation and movement smoothness metrics, have an even more defined baseline for each of the trials to be as consistent as possible (e.g.

well defined initial position), and extend the proposed framework to post-stroke populations.

5 Conclusion

The objective of this study was to quantify inter-limb asymmetry during representative self-feeding activities using an OpenSim-based kinematic analysis framework. The results demonstrate that measurable asymmetries exist even in healthy young adults, with

shoulder abduction/adduction consistently exhibiting the largest inter-limb differences, supporting the hypothesis that dominant and non-dominant upper limbs do not necessarily perform functional tasks in an identical manner. Task complexity also influenced asymmetry magnitude, with cutting and bottle opening producing greater asymmetries than spoon and drinking tasks.

The present findings establish a preliminary baseline of healthy upper-limb asymmetry during self-feeding activities. Such baseline data may be useful for future comparisons with neurological populations, particularly post-stroke patients, in whom altered movement patterns and compensatory strategies are frequently observed.

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